← Back **The Habitability of Mars**
Graded Quiz • 30 min

Due Mar 18, 11:59 PM CET

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Go to next item

1. Which of the following conditions must be met for a planet to be habitable? — 1 / 1 point

- [x] It must have nutrients for life
  - Correct: Nutrients are required to grow and reproduce
- [x] It must have energy supplies for life
  - Correct: Energy is needed to grow and reproduce
- [ ] It must be near freezing
- [x] It must have a source of liquid water
  - Correct: Liquid water is required for life to do biochemistry
- [ ] It must be a small planet
- [ ] It must have a source of potassium

2. What is the classical 'habitable zone'? — 1 / 1 point

- ( ) The region around a star in which microbes can exist
- (•) The region around a star in which liquid water is stable on a planetary surface
- ( ) The region around a star in which there is enough light for photosynthesis
- ( ) The region around a star in which there is always life

Correct: This is correct, the habitable zone is calculated as the distance from a star at which liquid water would be stable.

3. Liquid water can exist outside the classical habitable zone because — 1 / 1 point

- ( ) Water can be liquid wherever there is life.
- ( ) The habitable zone can flip into different places quickly and at different times
- ( ) Water has different physical properties in different places
- (•) Tidal buckling, such as in the Jovian moon Europa, can cause liquid water to form far away from the habitable zone

Correct: Even though the planet is far away from the habitable zone, the interior of a moon like Europa can still have liquid water

4. The habitable zone around an F star is further from its star than our Sun because — 1 / 1 point

- ( ) F stars are short lived so the habitable zone must be further away
- ( ) F stars are cooler and so the habitable zone is further away
- ( ) F stars have more intense ultraviolet radiation so the habitable zone must be further away
- (•) F stars are hotter and so the habitable zone is further away

Correct: Our Sun is a G type star which is cooler than an F star. F stars have their habitable zone further away

5. The Moon has been said to affect the habitability of the Earth by — 1 / 1 point

- ( ) Reflecting more light and helping plants to photosyntheise
- ( ) Preventing asteroids from hitting the Earth
- ( ) Destabilizing the tilt of the planet
- ( ) Providing nutrients
- (•) Stabilizing the tilt of the planet

Correct: The Moon is indeed thought to stabilize the axis of the Earth and some people think that this makes the climate more stable than it otherwise would be.

6. The Viking Lander Biology experiments were designed to detect life. Today, what do scientists consider that it demonstrated? — 1 / 1 point

- ( ) That is showed the indisputable presence of life
- ( ) This it showed the presence of certain types of microbes in the Martian soil
- (•) That it showed the presence of a reactive surface chemistry
- ( ) That it showed the presence of large quantities of organics in the soil

Correct: The experiments are now generally interpreted as being the result of oxidants in the martian soil (such as perchlorates).

7. Astronomers in the early twentieth century thought they had seen canals on Mars. What are these now believed to be? — 1 / 1 point

- (•) A processing trick of the human brain in visualising large natural features on Mars
- ( ) Changes in the natural cover of vegetation on Mars
- ( ) Seasonally changing deserts on Mars
- ( ) Artificial canals built by a long-dead civilization

Correct: It is thought that astronomers such as Lowell and others saw the natural deserts and other variable features on Mars and their minds made linking lines between them - an optical processing trick of the human brain.

8. Grains of the iron oxide, magnetite, in a meteorite, are evidence for what? — 1 / 1 point

- ( ) Only evidence for chemical processes
- ( ) Evidence for decayed remains of animals
- (•) Possible non-biological or biological processes, depending on their shape
- ( ) Undeniable evidence of the activity of magnetotactic bacteria

Correct: Yes, magnetite crystals can be produced by both biological and chemical reactions.

9. In 1996, scientists reported finding possible evidence of fossil life in a Martian meteorite. Which one of the following was suggested as evidence for life? — 1 / 1 point

- ( ) Layers of cell colonies
- ( ) Amino acids
- ( ) Remains of cell membranes
- (•) Polyaromatic hydrocarbons

Correct: PAHs can be produced chemically, but their association with carbonate globules in specific locations in ALH84001 was claimed as evidence for ancient martian life.

10. Life needs which of the following to persist on a planetary surface? — 1 / 1 point

- [ ] A source of calcium
- [ ] A source of methane
- [x] A source of energy
  - Correct: Life needs energy to grow and reproduce
- [ ] A source of heat
- [x] A source of water
  - Correct: Life needs water to carry out biochemical reactions

11. In what period of Martian history is it thought that some of the earliest valley networks and lakes were formed? — 1 / 1 point

- (•) Noachian
- ( ) Paleozoic
- ( ) Edinburghian
- ( ) Amazonian

Correct: The Noachian is the period of Mars history about 4.1 to 3.7 billion years ago when many of Mars' liquid water features such as lakes and rivers seemed to have formed

12. Alternating stripes in the intensity of magnetic fields observed in rocks in southern hemisphere of Mars are thought to indicate what? — 1 / 1 point

- (•) Early plate tectonics on Mars
- ( ) Ancient water on Mars
- ( ) Ancient life on Mars
- ( ) Giant meteorite impacts

Correct: The strips are thought to represent rock flowing across the surface of Mars and as it cooled 'trapping' the Martian magnetic field

13. Which of the following conditions are thought to make the surface of Mars uninhabitable today? — 1 / 1 point

- [x] High radiation exposures
  - Correct: High radiation (including ultraviolet and ionizing radiation) would add to the deleterious conditions on the surface, particularly in combination with desiccation as organisms cannot repair themselves.
- [x] Lack of liquid water
  - Correct: The lack of liquid water is crucial because it prevents life from doing biochemical reactions
- [ ] Lack of phosphate
- [ ] Low concentrations of organics
- [ ] High oxygen levels

14. Today, where do people think might be the best place to look for extant life on Mars? — 1 / 1 point

- ( ) On the surface of the polar caps
- ( ) In the atmosphere
- ( ) In the core of Mars
- (•) In the subsurface

Correct: In the subsurface there may be liquid water and life would be protected from the harsh conditions at the surface

15. Planetary protection is concerned with protecting science on other planets. What is the name given to the problem of transferring life to extraterrestrial environments? — 0 / 1 point

- ( ) Forward contamination
- ( ) Drop contamination
- (•) Back contamination
- ( ) Advance contamination

Incorrect: Back contamination refers to extreterrestrial life being accidentally brought back to the Earth

16. One way to sterilize a spacecraft prior to its launch is to — 1 / 1 point

- (•) Sterilize it with heat
- ( ) Put it into a microwave oven
- ( ) Leave it for a long time for the microbes to die
- ( ) Wash it with soap

Correct: This is one way and was used to sterilize the Viking Landers sent to Mars in the 1970s.

17. Evidence of liquid water on the present-day surface of Mars might include — 1 / 1 point

- ( ) Fast flowing rivers observed in the northern hemisphere
- ( ) Rain storms during the summer months
- (•) Seeps of briny fluids
- ( ) Lakes of liquid water

Correct: Seeps that seem to change seasonally have been observed using the Mars Reconnaissance Orbiter. They are thought to represent possible brines seeping from the subsurface.

18. The hypothetical mechanism by which life might be transferred from one planet to another has been referred to as — 1 / 1 point

- ( ) Rocky life
- ( ) Litholife
- (•) Panspermia
- ( ) Biogeography

Correct: Correct

19. What is the strongest evidence on the Earth that rocky material is transferred between planets? — 1 / 1 point

- (•) Because Martian meteorites have been recovered on the surface of the Earth
- ( ) Because the distances between planets are small
- ( ) Because all planets eject rocky material
- ( ) Because all planets are always breaking apart

Correct: Martian meteorites have been found on the Earth. Their composition and small pockets of gas within them that have an atmospheric and isotopic composition similar to the Martian atmosphere are evidence they are from Mars

20. Which of the following conditions would have to be met for life to be transferred between a planet? — 1 / 1 point

- [x] Life must survive in the rock during atmospheric entry
  - Correct: Life must survive inside the rock as it enters the atmosphere of the host planet.
- [x] Life in rocks must survive transfer in space
  - Correct: Life must survive the radiation environment of space during its transfer from one planet to another.
- [ ] Life must always have a source of nutrients during its transfer from one planet to another
- [ ] Life must be kept frozen throughout the journey
- [x] Rocks must be ejected from a planet at low enough shock pressure for life to survive
  - Correct: If the shock of an asteroid or comet is too large then life inside the rock will be killed.
- [ ] Life in rocks must always have a source of liquid water